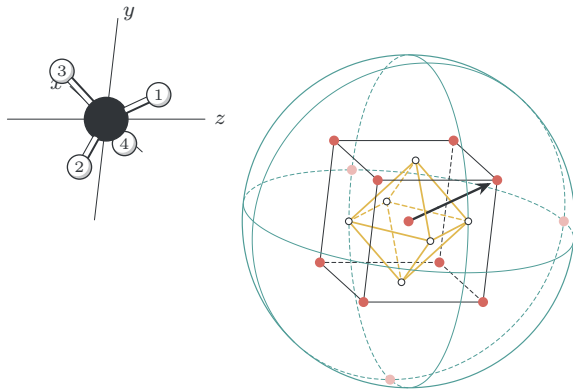
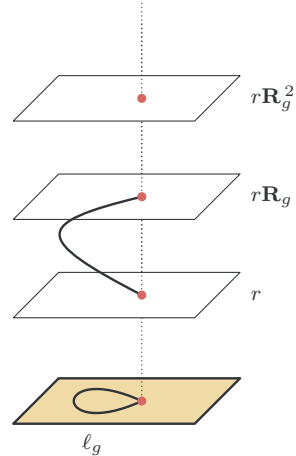


Methane, CH_4 : the reference $\mathbf{X}_0$ in its body frame,
with the C_2 axes x, y, z through the midpoints of opposite edges.



The orientation space $SO(3)$ as a ball,
direction the axis and distance the angle:
the twelve rotations of T ,
the cell $\mathcal{U}$ at $q = 0$ in gold, and the lift
 $r \rightarrow r\mathbf{R}_g$ of the third-turn loop.



The covering $SO(3) \rightarrow SO(3)/T$,
as in Figure 11: three of the twelve
sheets, one per rotation in T ; the loop ℓ_g
in the base, and its lift from sheet r to $r\mathbf{R}_g$.
A sees 1, ${}^1\text{E}$ sees ω , ${}^2\text{E}$ sees ω^2 ,
and T cycles x, y, z .